EXPERIMENT - 7

TO PREPARE AND COLLECT HYDROGEN GAS AND STUDY ITS PROPERTIES.

APPARATUS REQUIRED

a) Woulfe's Bottle
b) Thistle funnel
c) Gas Jar
d) Water through
e) Beehive shelf
f) Delivery tube
g) Corks with single bore
h) Test tubes

CHEMICAL REQUIRED

i. Zinc pieces
ii. Dilute H_2SO_4
iii. Litmus papers
iv. $FeCl_3$ solution
v. Acidified $K_2Cr_2O_7$ soln
vi. Acidified $KMnO_4$ soln

THEORY

Electropositive metals like Mg, Ca, Al, Zn etc. which are above hydrogen in the electrochemical series react with dilute mineral acids like HCl and H_2SO_4 to produce hydrogen gas.

In the laboratory, Hydrogen gas is prepared by reacting granulated zinc with dil. H_2SO_4

$$Zn(s) + dil.\ H_2SO_4 \rightarrow ZnSO_4(aq.) + H_2\uparrow$$

Ionic rxn: $Zn(s) + 2H^+(aq) \rightarrow Zn^{++}(aq) + H_2\uparrow$

In this reaction, Zn is oxidized Zn^{++} and H^+ is reduced to H_2.

The newly born hydrogen on metal surface during chemical reaction (i.e. in situ) is called nascent hydrogen.

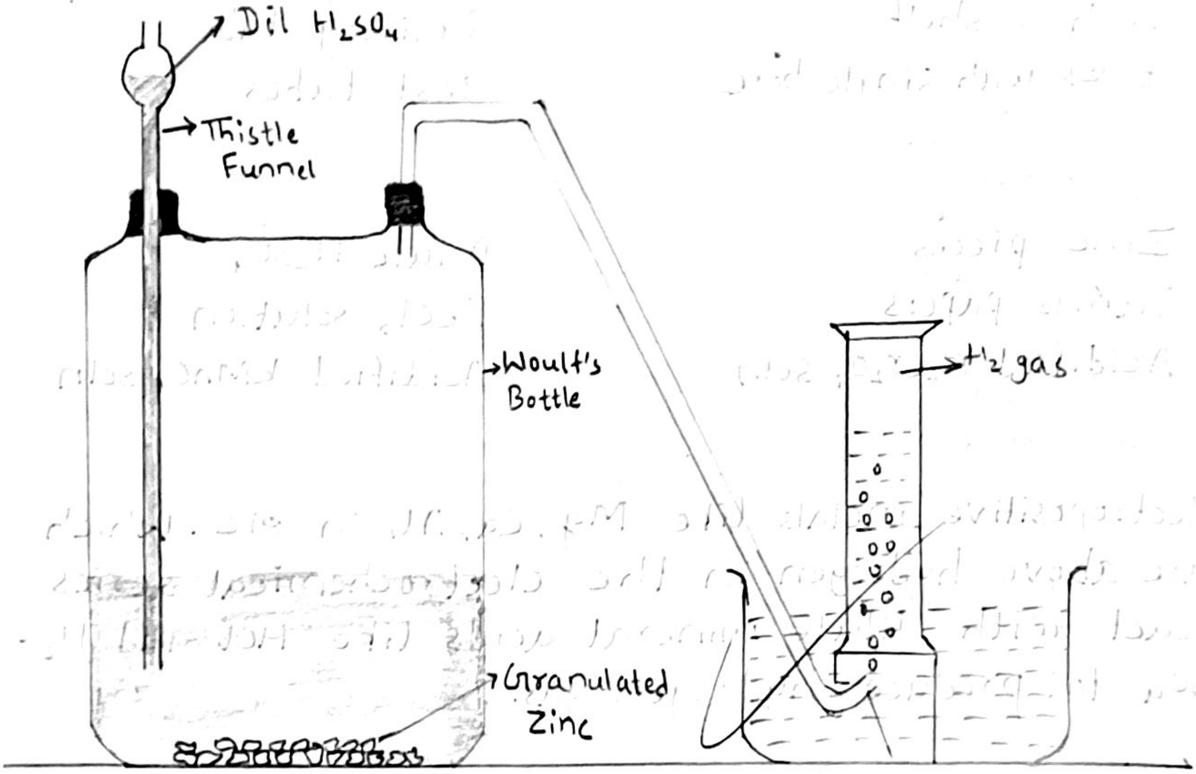

fig:- Laboratory preparation of hydrogen gas

Hydrogen gas is lighter than air and very slightly soluble in water. Therefore, it is collected by the downward displacement of water.

Pure zinc cannot be used for the preparation of hydrogen gas because the hydrogen gas produced during reaction is adsorbed on the surface of zinc metal there by preventing the contact of zinc with dilute mineral acid and the reaction will cease.

$Zn + Conc.\ 2H_2SO_4 \longrightarrow ZnSO_4 + 2H_2O + \underline{SO_2}\uparrow$

Concentrated sulphuric acid is also an oxidizing agent and oxidizes nascent hydrogen water and itself get reduced to SO_2 gas when treated with zinc.

PROCEDURE

Some piece of granulated zinc was added into a clean woulfe's bottle. A thistle funnel was adjusted in one opening and a delivery tube was fitted in another opening of the woulfe's bottle with a cork, and the apparatus was made perfectly air tight. Dilute H_2SO_4 was poured through the thistle funnel to cover the zinc piece in the bottle, and the tip of the thistle funnel was dipped in the acid. A reaction occurred with effervesce and the gas was collected in a gas jar by the downward displacement of water, as shown in figure.

OBSERVATION TABLE

S.N.	Experiments	Observations	Inferences
1.	Colour and odour test ~~Obser.~~ Colour and odour of the gas was observed	No color was seen and no odour was found of that gas.	Hydrogen gas is colorless and odorless.
2.	Litmus paper test Moist red and blue litmus papers were introduced in the jar filled with Hydrogen gas	No change in Color of litmus paper was seen.	Hydrogen gas is neutral
3.	Combustibility test A burning match stick or piece of paper was introduced in the jar of Hydrogen gas.	Burning matchstick entinguished but the gas was burnet at mouth with blue flame.	H_2 gas is combustible but not supporter of combustion.
4.	Solubility Test A test tube filled with H_2 gas was inverted over water in a water trough.	No increase in level of water was seen inside test tube	H_2 gas is almost insoluble in water.
5.	Pop Sound & density test		

	In a test tube filled with H_2 gas, a test tube was inverted mouth to mouth and upper tube was ignited	Gas in upper tube was burnet with pop sound.	H_2 gas is lighter than air. It is oxidized with O_2 of air and is reducing agent.
6.	Reaction with dil. $KMnO_4$		
a	H_2 gas was passed into acidified $KMnO_4$ solution	change in color was not seen.	Molecular Hydrogen can't reduce acidified $KMnO_4$ soln.
b	Small piece of Zn were added into above acidified soln of $KMnO_4$	Pink color of acidified $KMnO_4$ was discharged in few min.	[H] reduce acidified $KMnO_4$ into colorless $KMnO_4$
7.	Reaction with dil. $K_2Cr_2O_7$		
a)	H_2 gas was passed into $K_2Cr_2O_7$ acidified with dil. H_2SO_4	change in color was not seen.	H_2 can't reduce acidified $K_2Cr_2O_7$
b	Some piece of Zn were added into above acidified $K_2Cr_2O_7$	Orange color of $K_2Cr_2O_7$ was changed into light green.	[H] reduces acidified $K_2Cr_2O_7$ into light green color of $Cr_2(SO_4)_3$
8.	Rxn with acidified $FeCl_3$		
	a) Passed H_2 gas into $FeCl_3$ soln acidified with dil. HCl	No change in color	H_2 can't reduce $FeCl_3$ solution

b. Added some pieces of zn in above acidified solution.	Reddish yellow color of $FeCl_3$ changes to greenish yellow color.	[H] reduce above reaction in greenish yellow color ($FeCl_2$)

TEST REACTIONS

1. Action with acidified $KMnO_4$ soln

a) Reaction of acidified $KMnO_4$ with H_2 gas

$KMnO_4 + dil.\ H_2SO_4 + H_2 \longrightarrow$ No action

b) Reaction of acidified $KMnO_4$ with nascent hydrogen

$$2KMnO_4 + dil.\ 3H_2SO_4 + 10[H] \xrightarrow{Zn/H_2SO_4} K_2SO_4 + 2MnSO_4 + 8H_2O$$

(Pink) (colourless)

2. Action with acidified $K_2Cr_2O_7$ soln

a) Reaction of acidified $K_2Cr_2O_7$ soln with H_2 gas

$K_2Cr_2O_7 + dil.\ H_2SO_4 + H_2 \longrightarrow$ No action

b) Reaction of acidified $K_2Cr_2O_7$ soln with [H]

$$K_2Cr_2O_7 + dil.\ 4H_2SO_4 + 6[H] \xrightarrow{Zn/H_2SO_4} K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O$$

(Orange) (Green)

3. Action with acidified $FeCl_3$ soln

a) Reaction of acidified $FeCl_3$ soln with H_2 gas

$FeCl_3 + dil.\ HCl + H_2 \longrightarrow$ No action

b) Reaction of acidified $FeCl_3$ soln with nascent hydrogen

$$FeCl_3 + [H] \xrightarrow{Zn/HCl} FeCl_2 + HCl$$

(Reddish brown) (Light green)

RESULT

In this way, Hydrogen gas was prepared by reacting granulated zinc with dil. mineral acids and its properties are studied.

CONCLUSION

Hence from above experiment it is concluded that hydrogen gas can be prepared in laboratory by the action of reactive metals on mineral dil. acids.

PRECAUTIONS

1. Personal Protective equipments should be worn.
2. The Apparatus should be completely airtight.
3. Air present inside the apparatus should be removed completely by displacing it with the gas.
4. Gas jar with water should be free from air bubbles while inverted over the beehive shelf inside water trough.

VIVA-VOCE

1. **Why is copper metal not used to prepare hydrogen gas in lab?**
2. **Why is conc. H_2SO_4 cannot be used instead of dil. H_2SO_4?**
3. **Can we use nitric acid to prepare hydrogen gas?**
4. **Which form of hydrogen can act as most powerful reducing agent?**

1. Because Cu is less electropositive than hydrogen and lies below hydrogen in electrochemical series and cannot displace the H_2 gas from dilute mineral acids.

2. Conc. H_2SO_4 is good oxidizing agent and it oxidizes Zn and itself gets reduced to SO_2.

3. No, nitric acid is generally not used to prepare H_2, except very dil. acid with reactive metals like Mg or Mn.

4. Atomic hydrogen > Nascent hydrogen > Molecular hydrogen

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